

ADVANCED TRANSDISCIPLINARY RESEARCH METHODS

Exploring User Experience of AI-Initiated Interruptions in Everyday Voice AI Interactions

A SYNTHETIC MIXED METHODS STUDY

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Why this project?

Relevant to personal interest:

Prior work in voice AI interaction revealed a recurring UX friction point, AI interrupting at the worst possible moments

An understudied problem:

AI-initiated interruptions are disruptive yet rarely studied as a standalone phenomenon

Existing Framework

No to very limited validated model exists for how users actually experience these interruptions

This Project

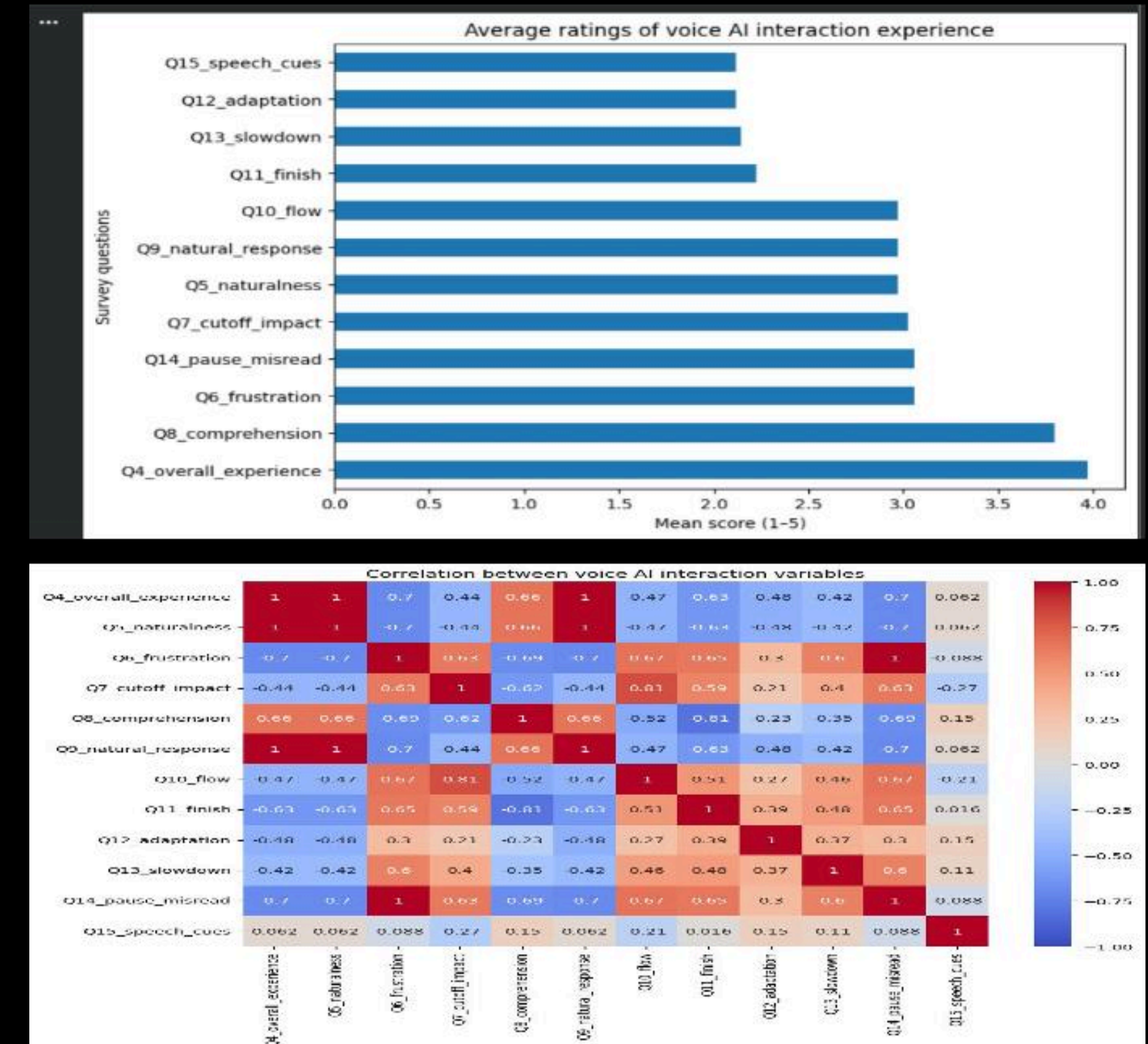
A systematic, mixed-methods exploration, not related to Masters project

Study Methods: Exploratory Sequential Design

Method 2 - Quantitative Phase

- 5 themes → 15-item Likert survey
- n=35 participants from Jain & Basu (2023) profiles
- Human-grounded synthetic simulation
- Analysis: Mean scores → Pearson correlation

Appendix | Survey Questions: https://docs.google.com/document/d/1u4_LCVz74u6FhCx_EjvRfzzxlcpLIPNJvxGqRxQdLtc/edit?usp=sharing



Results

01. Emotional Frustration

Being cut off mid-sentence → negative experience

02. Comprehension Failure

AI fails to understand like a human would

03. Flow Disruption

Can't finish sentences, constant interruption

04. Behavioral Adaptation

Users change HOW they speak to accommodate AI

05. Paraverbal Cue Misinterpretation

AI misreads pauses & breath as conversation end

73.2% adapt | $r = 1.0$

Users are resilient but burdened, frustration doesn't reduce with experience

Conclusion

- 🔄 Voice AI violates fundamental turn-taking norms (Sacks et al., 1974)
- ⚖️ Users adapt but absorb the communicative burden that AI cannot manage
- 📈 Frustration doesn't reduce with experience → a structural design problem, not a user problem
- 💡 **Fix: AI that distinguishes thinking pauses from genuine conversation endings**
- ✍️ Synthetic simulation is a valid pilot methodology for future UX research